Barath R

Embedded Systems & Firmware Engineer | Full-Stack & IoT Systems

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PROFESSIONAL SUMMARY

Third-year Electronics & Communication Engineering student and Embedded Firmware Engineer with hands-on proficiency across hardware design, bare-metal firmware, and software engineering. Experienced in building from silicon to cloud -- from register-level RISC-V assembly and RTOS firmware (FreeRTOS & Zephyr RTOS) to custom PCB/sensor bring-up, device drivers, full-stack IoT telemetry pipelines, and algorithmic problem solving. Seeking an Embedded Engineer role to architect high-performance, connected hardware and software systems.

EDUCATION

B.E. Electronics and Communication Engineering 2024 -- 2028 (Expected)
Rajalakshmi Institute of Technology, Chennai CGPA: 8.65/10
Currently in 3rd year

Higher Secondary Certificate (HSC) -- Class XII 2023 -- 2024
Montfort Matriculation Higher Secondary School, Chennai Percentage: 95%
Tamil Nadu State Board

Secondary School Leaving Certificate (SSLC) -- Class X 2021 -- 2022
Montfort Matriculation Higher Secondary School, Chennai Percentage: 94%
Tamil Nadu State Board

TECHNICAL SKILLS

Hardware & Embedded Systems: ESP32-C6 (RISC-V), ESP32, ESP32-CAM, Raspberry Pi 5, Arduino UNO, Altera DE2 FPGA, Bare-Metal MMIO, Oscilloscope (DSO), Logic Analyzer, JTAG/SWD, GDB, OpenOCD, CST Studio, Keysight ADS

Firmware & Low-Level Development: Bare-Metal C/C++, FreeRTOS, Zephyr RTOS, ESP-IDF (v6), RISC-V Assembly, Device Drivers, BSP, HAL, ISR, DMA, Timers, PWM, ADC, Low-Power Modes

PCB Design & EDA Tools: Printed Circuit Board (PCB) Design, KiCad, Proteus, EasyEDA, Schematic Capture, 2/4-Layer Layout, Routing, DRC/ERC, Gerber & BOM, Soldering & Board Bring-up

Hardware Protocols & Interfacing: I2C, SPI, UART/USART, CAN Bus, GPIO, PWM, ADC, MMIO, MQTT, LoRaWAN, ESP-NOW, WebSockets, HTTP/REST

Programming Languages: C, Embedded C, C++, Python, JavaScript (ES6+), TypeScript, Java, SQL, RISC-V Assembly

Software & Web Development: React.js, Next.js, Node.js, Express.js, FastAPI, Flask, WebAssembly (WASM), Web Workers, REST APIs

Cloud, Databases & DevOps: Supabase (PostgreSQL), Firebase, HiveMQ, ThingSpeak, AWS IoT Core, Google Cloud Platform, MongoDB, Docker, Git, GitHub Actions (CI/CD)

Core CS & Problem Solving: Data Structures & Algorithms (LeetCode streak), OOP, Concurrency, Multithreading, RTOS Architecture (FreeRTOS & Zephyr)

TECHNICAL PROJECTS

Embedded Systems & Hardware Engineering

Bare-Metal RISC-V Trap Handler & Context Switcher *ESP32-C6 / Bare-Metal*
Tech Stack: RISC-V Assembly, C, Direct MMIO, Hardware Timers, Interrupt Controller

- Architected a bare-metal trap handler and preemptive context switcher in assembly with 0-byte RTOS overhead, achieving $<$ 2.4 us context-switch latency across 32 registers
- Configured direct Memory-Mapped I/O (MMIO), machine timer interrupts (MTIE), and trap vectors (mtvec) for sub-microsecond hardware interrupt handling and register preservation

ESP-Drone -- Autonomous Flight Controller & Firmware *ESP32-C6 / Quadcopter Firmware*
Tech Stack: C/C++, ESP-IDF, FreeRTOS, IMU Sensor Fusion, PID Control, ESP-NOW / Wi-Fi, MCPWM

- Engineered real-time flight control firmware for an autonomous micro-quadcopter powered by the ESP32-C6 RISC-V MCU, executing a 500 Hz closed-loop attitude stabilization cycle with $<$ 1% loop jitter
- Implemented 6-DOF IMU sensor fusion with complementary filtering and quaternion attitude estimation, maintaining +-1 deg orientation accuracy and automated failsafe cutoffs
- Developed low-latency bidirectional telemetry via ESP-NOW ($<$ 5 ms latency) and high-resolution 20 kHz motor MCPWM driver pipelines

Digital Twin for Industrial Monitoring *5G-Integrated IoT System*
Tech Stack: ESP32 / Edge Node, Multi-Sensor Fusion, 5G Telemetry, Python, Real-Time Web Dashboard

- Architected an industrial digital twin connecting multi-sensor ESP32 edge nodes with 5G cellular telemetry, sustaining 99.8% data transmission uptime
- Engineered a sub-100 ms edge-to-cloud data ingestion pipeline and real-time dashboard for predictive thermal, vibration, and current anomaly detection

Dual-Axis Solar Tracker *Embedded Control System*

Tech Stack: Custom PCB Prototyping, Microcontroller, Servo Motors, LDR Sensor Array, Closed-Loop Control, C/C++

- Constructed a closed-loop dual-axis solar tracking system, increasing photovoltaic energy capture by 28% over static panels through analog LDR differential sensing
- Fabricated motor driver and voltage regulation circuits; programmed dual-servo tracking algorithms across 180 deg azimuth and 90 deg elevation with low-power sleep modes

Conformal Helmet-Integrated Antenna Array *RF & Defense Hardware*

Tech Stack: Microstrip Patch Array, Dual-Band UHF/L-Band, Flexible PCB Substrate, CST Microwave Studio, ADS, RF Shielding

- Engineered a flexible microstrip patch antenna array conformal to curved helmet geometry, achieving 45% profile reduction and S11 return loss $<$ -10 dB across UHF and L-band
- Simulated 3D radiation patterns and gain in CST Microwave Studio and Keysight ADS, integrating copper-clad polyimide substrates with high-isolation RF shielding

Software & Full-Stack Web Engineering

iCreatePDF *100% Client-Side Privacy PDF Suite*

Tech Stack: Next.js 16, TypeScript, WebAssembly (WASM), Web Workers, Tailwind CSS

- Architected a 100% client-side privacy-first PDF utility suite with 37+ tools, eliminating server processing and egress costs by executing in-memory via WebAssembly
- Leveraged Web Workers and WASM for parallel execution, processing 50 MB documents in $<$ 800 ms for compression, page re-ordering, and booklet imposition
- Deployed production web application at icreatepdf.online featuring offline PWA support, achieving 100/100 Google Lighthouse performance and accessibility scores

QRLoop *Dynamic QR & Scan Analytics Platform*

Tech Stack: React.js, Node.js, Express.js, Supabase (PostgreSQL), JWT, HTML5 Canvas

- Engineered a full-stack dynamic QR platform handling real-time destination redirection with $<$ 50 ms latency and 99.9% uptime using Node.js and Supabase
- Built real-time scan analytics capturing device telemetry, referrers, and geolocation with rate-limited JWT authentication and IP throttling
- Developed an HTML5 dynamic canvas rendering engine supporting SVG export, custom error-correction levels (L/M/Q/H), and brand styling

FlashESP *Web Serial Firmware Flasher & In-Browser IDE*

Tech Stack: Next.js, Web Serial API, esptool-js, Monaco Editor, Supabase, Tailwind CSS

- Built a driverless Web Serial ESP32 firmware flasher achieving 921,600 baud flashing speeds, reducing developer setup time from 15 minutes to under 30 seconds
- Integrated Monaco Editor with C/C++ syntax highlighting, automatic baud rate detection, and interactive ANSI terminal logging for live device telemetry

Energize 2026 Hackathon Portal *Interactive 3D Web Application*

Tech Stack: React.js, TypeScript, Three.js, GSAP, Web Audio API, Vercel

- Engineered official symposium and hackathon portals handling 1,000+ registrations with interactive 3D WebGL scenes rendered via Three.js
- Optimized rendering pipeline using GSAP animation timelines and Web Audio API, sustaining steady 60 FPS performance across mobile and desktop browsers

TECHNICAL EXPERIENCE & LEADERSHIP

Executive Technical Organizer 2024 -- Present

IETE Students' Forum (ISF) Chennai, India

- Spearheaded technical hackathons and hands-on embedded workshops, mentoring 500+ student engineers across firmware development, microcontroller interfacing, and MCU bring-up
- Conceptualized, engineered, and brought up 10+ embedded and full-stack systems end-to-end, spanning custom 2/4-layer PCB layout, bare-metal firmware, and real-time cloud telemetry
- Architected open-source software tools (iCreatePDF, QRLoop, FlashESP), adopting WebAssembly, TypeScript, and modern API architectures with 100% test pass rate